STACKS, DERIVED CATEGORIES OF SPACES: SUPPORT-PROPER-OVER-BASE-PUSHFORWARD

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1354-1361 of sources/stacks-spaces-perfect/pinned-original-source.tex; proof span is lines 1363-1376, the complete author proof (12 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 5 of queue_spaceperfect.json (pb=660, stmt 1354-1361, proof 1363-1376).

Editorial note. stacks-spaces-perfect wave sp01 (2026-10-14): fresh adjudication accept per lane-s3/sp01_judgments; pool per Chief 12:4x unfreeze (reserve-pool workflow, six-check clean); spaces-perfect.tex 126 with-proof rows, 70 b107-occupied labels avoided, 56 free rows judged 32 accept / 24 reject.

Editorial note. gate_ok=true: raw-interval guard false positive (public-items.jsonl upstream-frame coords); precard_gate.py shift-corrected check cleared this label 2026-10-14 (gate_sp01.sh).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0CZI: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1248–1263. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0734: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1376–1376. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07UH: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1376–1376. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08AR: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1376–1376. Cited by the retained statement or proof but

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume f is locally of finite type and Y locally Noetherian. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module with support proper over Y. Then $R^p f_* \mathcal{F}$ is a coherent $\mathcal{O}_Y$-module for all $p \geq 0$.*

Proof. By Lemma 0CZI there exists a closed immersion $i : Z \to X$ with $g = f \circ i : Z \to Y$ proper and $\mathcal{F} = i_* \mathcal{G}$ for some coherent module $\mathcal{G}$ on Z. We see that $R^p g_* \mathcal{G}$ is coherent on S by Cohomology of Spaces, Lemma 08AR. On the other hand, $R^q i_* \mathcal{G} = 0$ for $q > 0$ (Cohomology of Spaces, Lemma 07UH). By Cohomology on Sites, Lemma 0734 we get $R^p f_* \mathcal{F} = R^p g_* \mathcal{G}$ and the lemma follows. □